

PRODUCT REQUIREMENTS DOCUMENT

Navi Mumbai House Price Predictor

An AI-Powered Real Estate Valuation Platform

Field	Details
Document Version	1.0
Status	Draft
Date	February 2026
Target Launch	Q4 2026
Product Owner	TBD
Stakeholders	Product, Engineering, Data Science, Business

1. Executive Summary

The Navi Mumbai House Price Predictor is a web and mobile platform that uses machine learning to provide accurate, real-time property valuations across Navi Mumbai's diverse micro-markets — including Kharghar, Vashi, Belapur, Nerul, Panvel, Airoli, and surrounding nodes. The platform targets homebuyers, sellers, investors, and real estate agents seeking transparent, data-driven pricing insights in one of India's fastest-growing metropolitan regions.

The product will ingest historical transaction data, macroeconomic indicators, infrastructure developments (e.g., Navi Mumbai International Airport, Metro Line 1), and hyperlocal amenity data to generate price predictions with confidence intervals. Monetization will be through subscription tiers, lead generation for brokers, and an API for institutional clients.

2. Problem Statement

2.1 Market Context

Navi Mumbai's real estate market is characterized by significant price opacity. Buyers frequently overpay or underpay due to:

- Fragmented listings across platforms with inconsistent pricing
- Lack of hyperlocal data — prices vary drastically even within the same node
- Agent-led information asymmetry

- Rapid price shifts driven by infrastructure announcements (e.g., NMIA, Trans-Harbour Link, Metro expansion)

2.2 User Pain Points

User Segment	Pain Point	Impact
Homebuyer	Cannot verify if listed price is fair	Overpays or loses good deals
Seller	Uncertain about correct listing price	Underprices or property stays unsold
Investor	No ROI forecasting tool	Poor allocation decisions
Broker/Agent	Spends time on manual CMAs	Low productivity
Bank/NBFC	Manual property valuation for loans	Slow disbursement, risk exposure

3. Goals & Success Metrics

3.1 Product Goals

- Provide property valuations within $\pm 8\%$ of actual transaction price for 80% of predictions
- Cover all major CIDCO nodes of Navi Mumbai at launch
- Reduce time to price decision for buyers and sellers from days to minutes
- Build the most comprehensive real estate dataset for Navi Mumbai

3.2 Key Success Metrics (KPIs)

Metric	Target (6 months)	Target (12 months)
Prediction MAE (Mean Absolute Error)	< 10%	< 8%
Monthly Active Users	5,000	25,000
Registered Properties in DB	50,000	2,00,000
Paid Subscribers	500	3,000
API Partners	3	15
App Store Rating	4.0+	4.3+

4. Target Users & Personas

Persona 1: Rohan — The First-Time Homebuyer

Age: 31 | Occupation: IT professional in Mahape | Budget: ₹80L–₹1.2Cr

Goals: Buy a 2BHK in Kharghar or Nerul near metro. Verify that asking price is fair. Understand price trends for the next 2 years.

Frustrations: Overwhelmed by conflicting broker quotes. Doesn't know if prices will fall after NMIA opens.

Persona 2: Priya — The Real Estate Investor

Age: 44 | **Occupation:** Business owner | **Portfolio:** 3 properties in Panvel and Ulwe

Goals: Identify undervalued micro-markets, forecast rental yield vs capital appreciation, track infrastructure-driven price movements.

Frustrations: No single platform gives Navi Mumbai-specific analytics. Generic portals lack depth for serious investors.

Persona 3: Suresh — The Registered Broker

Age: 38 | **Occupation:** Independent broker, 12 years in Navi Mumbai

Goals: Generate automated Comparative Market Analyses (CMAs) for clients, get leads from serious buyers, position himself as a data-driven professional.

Frustrations: Manual price research takes hours. Clients distrust gut-feel pricing.

5. Feature Requirements

5.1 Core Features (MVP — Phase 1)

F1: Property Price Estimator

Users input property details and receive an instant price estimate with confidence range.

Input Parameter	Type	Required
Node/Locality (e.g., Kharghar, Sector 12)	Dropdown + autocomplete	Yes
Property Type (Flat, Villa, Plot, Row House)	Dropdown	Yes
Carpet Area (sq ft)	Numeric field	Yes
Floor Number	Numeric field	Yes
Total Floors in Building	Numeric field	Yes
Age of Property (years)	Slider / Year picker	Yes
Amenities (parking, gym, pool, etc.)	Multi-select checklist	No
Furnishing Status	Dropdown	No

Input Parameter	Type	Required
Facing Direction	Dropdown	No

Output: Estimated price (₹ and ₹/sq ft), confidence band (low–mid–high), price per sqft benchmark for locality, comparable recent sales.

F2: Price Trend Dashboard

- Historical price trends by node, BHK type, and property age (past 5 years)
- Quarter-on-quarter and year-on-year price change visualization
- Demand heatmap across Navi Mumbai nodes
- Rental yield estimates by locality

F3: Infrastructure Impact Tracker

- Map overlay of upcoming and completed infrastructure: NMIA, Metro Line 1, Trans-Harbour Link, Ring Road
- Price sensitivity analysis: how proximity to infrastructure correlates with price movement
- Notification alerts when new infrastructure is announced near a saved property

F4: Saved Properties & Portfolio

- Users can save property searches and estimates
- Track price changes on saved listings over time
- Export estimates as PDF reports

5.2 Advanced Features (Phase 2)

F5: ROI Forecasting Tool

- 2–5 year price appreciation forecast using ML models + scenario analysis (base, bull, bear)
- Rental yield calculator integrated with current rental market data
- Investment comparison: which node offers better returns

F6: Automated CMA for Brokers

- Broker-facing tool to generate professional Comparative Market Analysis PDF in under 2 minutes
- White-label PDF option with broker branding
- Shareable CMA link for clients

F7: API Access for Institutional Clients

- REST API for banks, NBFCs, and prop-tech platforms
- Bulk valuation endpoint (up to 1,000 properties per request)
- Webhook for price change notifications

F8: Conversational Price Assistant

- AI chatbot that answers natural language queries: 'What's a fair price for a 3BHK in Nerul near the station?'
- Powered by LLM integrated with live pricing data

6. ML Model Requirements

6.1 Data Sources

Data Source	Type	Update Frequency
IGRS Maharashtra (Stamp Duty Records)	Transaction prices	Monthly
MahaRERA	Project registrations, builder data	Weekly
99acres, MagicBricks, Housing.com	Listed prices, inventory	Daily
NMMC / CIDCO Development Plans	Infrastructure data	Quarterly
OpenStreetMap + Google Maps API	Amenities, transit proximity	Monthly
RBI / NHB Housing Index	Macroeconomic indicators	Quarterly
News APIs (Economic Times, etc.)	Sentiment & announcement data	Daily

6.2 Model Architecture

- Primary model: Gradient Boosted Trees (XGBoost / LightGBM) for structured tabular data
- Geospatial features: Haversine distances to key landmarks, node centroid embeddings
- Time-series component: ARIMA / LSTM for trend forecasting
- Ensemble: Weighted average of multiple models with uncertainty quantification
- Retraining cadence: Monthly full retrain; weekly incremental update

6.3 Model Validation

- Train/validation/test split: 70/15/15 by time (not random) to prevent leakage
- Primary metric: MAPE (Mean Absolute Percentage Error) — target < 8%
- Secondary metrics: R², P10/P90 coverage (confidence interval calibration)
- Fairness audit: ensure no systematic bias by locality or property type

7. Technical Architecture

7.1 Technology Stack

Layer	Technology	Rationale
Frontend (Web)	React.js + TypeScript	Component reusability, SEO via Next.js
Mobile	React Native	Cross-platform iOS + Android
Backend API	FastAPI (Python)	ML-native, async, high performance
ML Platform	MLflow + AWS SageMaker	Experiment tracking, scalable serving
Database	PostgreSQL + PostGIS	Geospatial queries on property data
Cache	Redis	Fast repeated lookups for price estimates
Data Pipeline	Apache Airflow + dbt	Orchestrated ETL from multiple sources
Maps	Mapbox GL JS	Custom styled maps for Mumbai
Auth	Auth0	OAuth2, social login
Infrastructure	AWS (Mumbai region ap-south-1)	Data residency compliance

7.2 Non-Functional Requirements

Requirement	Specification
API Response Time (Price Estimate)	< 2 seconds (P95)
System Uptime	99.5% monthly SLA
Concurrent Users	Support 5,000 concurrent at launch
Data Encryption	AES-256 at rest; TLS 1.3 in transit
GDPR/IT Act Compliance	User data deletion within 30 days of request
Mobile App Size	< 30 MB download size
Accessibility	WCAG 2.1 AA compliance

8. Monetization Strategy

Tier	Price	Target Segment	Key Features
Free	₹0/month	Casual buyers	5 estimates/month, basic trends
Homebuyer Pro	₹299/month	Active buyers/sellers	Unlimited estimates, ROI tool, PDF export
Broker Plus	₹999/month	Agents & brokers	CMA tool, white-label reports, lead gen
Enterprise API	₹25,000+/month	Banks, NBFCs, prop-tech	API access, bulk valuation, SLA

Additional revenue: Featured listings for builders/developers (₹5,000–₹50,000/month), data licensing to research firms.

9. Risks & Mitigations

Risk	Probability	Impact	Mitigation
Data quality from IGRS — incomplete/delayed records	High	High	Supplement with scraped listing data; flag low-confidence estimates
Model accuracy degrades after infrastructure shock	Medium	High	Retrain model within 2 weeks of major announcement; manual override option
Regulatory scrutiny on property valuations	Low	High	Add clear disclaimer: estimates are not official valuations; partner with RERA-registered valuers
Competition from 99acres, MagicBricks launching similar features	High	Medium	Focus on Navi Mumbai depth; faster retraining cadence; broker loyalty program
User trust in AI-generated prices	Medium	Medium	Show comparable transactions as evidence; confidence intervals; explain model logic

10. Go-to-Market & Roadmap

Phase	Timeline	Deliverables
Phase 0: Data & Foundation	Months 1–3	Data pipeline live, ML baseline model (MAPE < 12%), DB with 50K+ transactions
Phase 1: MVP Launch	Months 4–6	Web app + Android app, F1–F4 features, Free + Pro tiers, 5K MAU target

Phase	Timeline	Deliverables
Phase 2: Growth	Months 7–9	iOS app, F5–F6 (ROI + CMA), Broker Plus tier, 15K MAU target
Phase 3: Enterprise	Months 10–12	API launch (F7), AI chatbot (F8), bank partnerships, 25K MAU target

11. Open Questions & Dependencies

- Will IGRS data be accessible via official API or will scraping be required? (Legal review needed)
- Should the platform cover rental market pricing in Phase 1 or defer to Phase 2?
- MahaRERA partnership — can we get preferential data access as a registered prop-tech platform?
- Localization: Will the app need Marathi and Hindi UI support at launch?
- Should price estimates include GST and stamp duty calculations?

12. Appendix

12.1 Navi Mumbai Node Coverage (Phase 1)

Vashi, Belapur (CBD Belapur), Kharghar, Nerul, Airoli, Ghansoli, Kopar Khairane, Turbhe, Sanpada, Juinagar, Seawoods–Darave, Panvel, New Panvel, Ulwe, Dronagiri

12.2 Glossary

Term	Definition
CIDCO	City and Industrial Development Corporation — the planning authority for Navi Mumbai
IGRS	Inspector General of Registration and Stamps — publishes official property transaction records
MahaRERA	Maharashtra Real Estate Regulatory Authority — registers real estate projects
NMIA	Navi Mumbai International Airport — major infrastructure project impacting Ulwe, Dronagiri, Panvel areas
CMA	Comparative Market Analysis — a report comparing similar recent property sales to estimate value
MAPE	Mean Absolute Percentage Error — primary model accuracy metric
MAU	Monthly Active Users

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